

A Logical Argument Toward the Strong Goldbach Conjecture

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Let me be clear: this is not a formal proof. It's an argument. But I believe it's strong, structured, and might reveal something deeper behind the Strong Goldbach Conjecture.

Main Idea

If an even number can be written as

$$2n = P_1 + P_2,$$

where both P_1 and P_2 are prime numbers, then the next even number $2n + 2$ can also be written as a sum of two primes by logically manipulating the previous pair.

Here's the trick:

If

$$2n = P_1 + P_2,$$

then:

$$2n + 2 = (P_1 + 2k) + (P_2 + 2 - 2k),$$

or alternatively:

$$2n + 2 = (P_1 - 2k) + (P_2 + 2 + 2k),$$

where k is some integer such that both new values are prime. The key here is: we redistribute the $+2$ and the shift ($\pm 2k$) over the pair in a way that preserves the even sum and tries to land again on a pair of primes.

Basically, you explore a set of transformations like:

$$(P_1 \pm 2k, P_2 \pm (2 \mp 2k))$$

until both are primes. That's the core method.

Why This Matters

The difference between primes is almost always even (except between 2 and 3), so playing with even shifts doesn't break the logic.

People worry that primes get too sparse. But that fear is wrong — the gaps between primes grow slower than the primes themselves.

So even as numbers get big, you still have enough primes to play with. The space between them is enough for the $+2k/-2k+2$ to work.

Also, let's talk about something else:

If you had two consecutive primes p and q , that would mean the prime gap is massive. That's impossible because the gap between consecutive primes can't grow that fast — it's way smaller than the growth of the primes themselves.

Now think: if P_1 and P_2 were the only primes in that region, then $2n$ would have no possible sum of primes. That would be a direct counterexample to Goldbach. But that never happens.

Conclusion

Even though this isn't a full-on proof, it's a structural framework. You take a valid Goldbach pair and you shift it intelligently — by adjusting both primes symmetrically and embedding the $+2$ into one of the terms. Because the primes are dense enough and structured well enough, you'll always find a new valid pair for the next even number.

It's not just faith — it's logic, wrapped around the way primes behave.

– *Amine*